

Quantitative typological analysis of Latin-script languages using n -gram models^{*}

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Abstract

Typological linguistics is increasingly shifting from manual data analysis towards algorithms based on processing massive text corpora. This paper presents an approach based on the analysis of raw corpora at the character n -gram level. The aim of this study was to investigate whether simple clustering algorithms are capable of reflecting the natural division of languages into language families. The analysis involved 32 language corpora, each consisting of the 15,000 most frequent words extracted from the OpenSubtitles database. Using the Word2Vec model, character 3-grams were embedded into a high-dimensional vector space. Based on these embeddings, centroids for individual words, and then for entire languages, were computed. Following the language embedding, clustering was performed using the Affinity Propagation algorithm, utilizing cosine similarity between language pairs. The clustering process resulted in the formation of six multi-member clusters, which were subsequently subjected to quantitative morphological analysis to determine their key characteristics, such as 2-gram entropy, average maximum consonant cluster length, vowel percentage, and the identification of the most frequent affixes. A comparison with the linguistic literature confirmed that the model accurately reconstructed real language groups, including Germanic, Romance, Slavic, and Austronesian families. The obtained results demonstrate that models based on simple n -gram analysis are capable of capturing orthographic, lexical, and morphological differences across languages of diverse origins.

Keywords

Latin-script languages, morphological analysis, affinity propagation, Word2Vec, AP, language clustering, affixes, n -grams

1. Introduction

The problem of language classification dates back to antiquity; for instance, the Romans believed that Latin was directly derived from Greek [1]. The first attempts at a rigorous division of languages appeared in the 14th century, notably through Dante Alighieri, who, in his treatise *De vulgari eloquentia*, classified Romance dialects into three groups based on the word for *yes*: the *oil* group, *oc* group, and *si* group [2]. In 1599, the Dutch scholar Joseph Scaliger categorized languages into four major families based on the word for *God*: Romance (*Deus*), Greek (*Theos*), Germanic (*Godt*), and Slavic (*Bog*) [3].

Nowadays, language clustering is shifting from manual analysis of individual words towards computational methods and the processing of massive language corpora. One of the dynamically developing fields dealing with language analysis is NLP (*Natural Language Processing*), in which machine learning algorithms are used for understanding, generating, and classifying text. One of

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the methods for language clustering is determining the average Levenshtein distance between corresponding words, as in the work of M. Serva and F. Petroni *Indo-European languages tree by Levenshtein distance* [4]; the algorithm successfully classified the languages, but this approach requires manual compilation of dictionaries because the algorithm operates on pairs of corresponding words. Another commonly used method for grouping languages is the k -means algorithm; however, the main disadvantage of this solution is the fact that fixing the number of clusters k a priori significantly affects the clustering result, which often allows for the artificial achievement of assumed results [5]. Modern alternatives are transformer models, such as mBERT, but their high computational complexity and focus on semantic representation mean they can blur subtle morphological features [6]. P. Bojanowski, in the article *Enriching word vectors with subword information*, shows that vectorization at the character n -gram level allows algorithms to capture word morphology [7].

In this paper, a combination of the Word2Vec [8] and Affinity Propagation [9] algorithms is proposed to create a fully automated and unsupervised clustering and typological analysis algorithm. The aim of the study was to verify whether a simple model based on character 3-gram vectorization can reconstruct linguistic divisions. For this purpose, a centroid representation for 32 languages was used, for which the research material consisted of corpora from the OpenSubtitles database [10]. The combination of Word2Vec vectorization with the Affinity Propagation algorithm allowed for the identification of natural language groups based on purely statistical text distribution.

2. Methodology

Thirty-two language corpora are subjected to analysis, each consisting of exactly 15,000 of the most frequent words. The study is based on raw text forms, which allows for the capturing of morphological patterns of individual languages. The scope is limited to 15,000 words in order to filter out statistical noise. The foundation of the research material consists of frequency lists obtained from the OpenSubtitles database [10]. The use of data based on film dialogue transcriptions allows for a faithful representation of natural, everyday speech.

Each corpus is subjected to a normalization process, which involves the removal of digits, white spaces, and punctuation marks. During preprocessing, # characters are added to the beginning and end of each word to mark word boundaries. Subsequently, every word from the analyzed corpora is divided into 3-grams, which are collected into a common training set D .

The training set D is embedded using the Word2Vec model [8]. The model is trained utilizing the Skip-gram architecture. The vectors are embedded into a 300-dimensional real vector space $\mathbb{R}^{300}$, with a context window size of 3 character tokens. To prevent underfitting, the model is trained for 60 epochs on the corpus, maintaining a minimum n -gram frequency threshold of 1.

To embed entire words w_i into the space $\mathbb{R}^{300}$, we calculate the centroids of all 3-grams comprising the word w_i . The centroid v_{w_i} is determined as the arithmetic mean of the vectors corresponding to the 3-grams $g \in F_3(w_i)$, where $F_n(w_i)$ denotes the multiset of all n -grams of word w_i , and v_g is the vector corresponding to the 3-gram g :

$$v_{w_i} = \frac{1}{|F_3(w_i)|} \sum_{g \in F_3(w_i)} v_g. \quad (1)$$

Analogously, to embed the entire language $\mathcal{L}$ in the $\mathbb{R}^{300}$ vector space, the centroid of all vectors v_{w_i} corresponding to words $w_i \in \mathcal{L}$ is determined:

$$v_{\mathcal{L}} = \frac{1}{|\mathcal{L}|} \sum_{w_i \in \mathcal{L}} v_{w_i}. \quad (2)$$

Averaging the entire language into a single vector is a deliberate procedure. It is intended to demonstrate that even with maximum numerical simplification and such a significant loss of data, the model is still capable of capturing typological differences.

The next step is to determine the similarity between languages. Let $\mathcal{C}$ denote the set of all 32 language corpora. For each pair of languages $(\mathcal{L}_1, \mathcal{L}_2) \in \mathcal{C} \times \mathcal{C}$, the cosine similarity S_C is determined by the formula:

$$S_C(\mathcal{L}_1, \mathcal{L}_2) = \frac{v_{\mathcal{L}_1} \cdot v_{\mathcal{L}_2}}{\|v_{\mathcal{L}_1}\| \|v_{\mathcal{L}_2}\|}. \quad (3)$$

All cosine similarities are collected in a matrix $\mathbf{M}$ of dimensions $|\mathcal{C}| \times |\mathcal{C}|$, where each i -th row and each j -th column correspond to languages $\mathcal{L}_i, \mathcal{L}_j \in \mathcal{C}$, respectively, where $i, j \in \{1, 2, \dots, |\mathcal{C}|\}$:

$$\mathbf{M} = [m_{ij}]_{|\mathcal{C}| \times |\mathcal{C}|}, \quad m_{ij} = S_C(\mathcal{L}_i, \mathcal{L}_j). \quad (4)$$

After the similarities S_C between languages have been determined, we can proceed to their clustering into language groups. For this purpose, the Affinity Propagation algorithm [9] is used. Unlike classical clustering methods, this algorithm does not require the number of clusters to be defined a priori. The clustering process is based on message passing between languages until an optimal set of exemplars representing individual language subgroups is identified. The preference parameter value p is set to $p = 0.9$. For lower values of p , the algorithm exhibited excessive fragmentation, creating numerous single-item clusters; the value $p = 0.9$ was found to be the model's stabilization point.

After extracting the clusters $\mathcal{G}$, we can proceed to perform a quantitative morphological characterization. Single-element clusters, which form isolated languages, are excluded from further analysis. Additionally, based on the distribution of boundary 3-grams (containing word boundary markers #), sets of the most frequent affixes are extracted.

Let $\mathcal{W}(\mathcal{G})$ denote the multiset of all words w such that $w \in \mathcal{L}$ and $\mathcal{L} \in \mathcal{G}$. The length of an individual word w is denoted as $|w|$. The average length of all words $w \in \mathcal{W}(\mathcal{G})$ is denoted as $\overline{|w|}$ and is expressed by the formula:

$$\overline{|w|} = \frac{1}{|\mathcal{W}(\mathcal{G})|} \sum_{w \in \mathcal{W}(\mathcal{G})} |w|. \quad (5)$$

The standard deviation of the lengths of words $w \in \mathcal{W}(\mathcal{G})$ is denoted as σ and is defined by the formula:

$$\sigma = \sqrt{\frac{1}{|\mathcal{W}(\mathcal{G})|} \sum_{w \in \mathcal{W}(\mathcal{G})} (|w| - \overline{|w|})^2}. \quad (6)$$

Let $|w|_V$ denote the number of vowels in word w ; the percentage of vowels in words $w \in \mathcal{W}(\mathcal{G})$ is denoted as P_V and is defined by the formula:

$$P_V = \frac{\sum_{w \in \mathcal{W}(\mathcal{G})} |w|_V}{\sum_{w \in \mathcal{W}(\mathcal{G})} |w|}. \quad (7)$$

Let $[w]_C$ denote the maximum length of a consonant cluster in word w ; the average maximum consonant cluster length in words $w \in \mathcal{W}(\mathcal{G})$ is denoted as $\overline{[w]}_C$ and is defined by the formula:

$$\overline{[w]}_C = \frac{1}{|\mathcal{W}(\mathcal{G})|} \sum_{w \in \mathcal{W}(\mathcal{G})} [w]_C. \quad (8)$$

Let $G_2 = \biguplus_{w_i \in \mathcal{W}(\mathcal{G})} F_2(w_i)$ denote the multiset of all 2-grams of words from the multiset $\mathcal{W}(\mathcal{G})$. The 2-gram entropy H_2 is determined according to the following pseudocode:

Algorithm 1: 2-gram entropy calculation

Data: G_2 – multiset of all 2-grams for a given group

Result: H_2 – 2-gram entropy expressed in bits

```

1  $t \leftarrow |G_2|$ ;
2  $C \leftarrow \{\}$ ;
3  $H_2 \leftarrow 0$ ;
4 foreach  $g \in G_2$  do
5   if  $g \in C$  then
6      $C[g] \leftarrow C[g] + 1$ ;
7   else
8      $C[g] \leftarrow 1$ ;
9 foreach  $c \in \text{values}(C)$  do
10   $p \leftarrow c/t$ ;
11  if  $p > 0$  then
12     $H_2 \leftarrow H_2 - p \cdot \log_2(p)$ ;
13 return  $H_2$ ;
```

The vectorization process was performed using the `Gensim` library in the Python environment, while the implementation of the Affinity Propagation algorithm from the `scikit-learn` package was used for clustering.

3. Experiments

3.1. Research Material

The material subjected to analysis consisted of 32 language corpora. For each language, 15,000 of the most frequent words in the Latin alphabet were selected, obtained from the OpenSubtitles database [10]. A compilation of all analyzed languages is collected in Table 1. The choice was

dictated by the aim to analyze natural, colloquial language occurring in everyday speech rather than more official literary language.

Table 1

List of the 32 languages included in the study.

Albanian	Basque	Bosnian	Catalan
Croatian	Czech	Danish	Dutch
English	Estonian	Finnish	French
Galician	German	Hungarian	Icelandic
Indonesian	Italian	Latvian	Lithuanian
Malay	Norwegian	Polish	Portuguese
Romanian	Serbian	Slovak	Slovenian
Spanish	Swedish	Turkish	Vietnamese

3.2. Language Similarity Network

The first result of the research was the language similarity matrix $\mathbf{M}$, which was presented using a heatmap in Figure 1. Matrix $\mathbf{M}$ was also visualized using an undirected graph, presented in Figure 2. The set of all languages $\mathcal{C}$ was adopted as the vertices of the graph, and an edge between languages $\mathcal{L}_1, \mathcal{L}_2$ exists if $S_C(\mathcal{L}_1, \mathcal{L}_2) \geq 0.96$.

Analysis of the generated graph revealed the formation of two distinct clusters, which coincide with existing language groups with high accuracy. On the left side of Figure 2, a large cluster can be observed, corresponding to Romance-Germanic languages, while on the right, a cluster corresponding to Balkan-Central European languages has formed. The graph also correctly demonstrates the similarity of Austronesian languages (Indonesian and Malay) and Balto-Finnic languages (Estonian and Finnish).

3.3. Typological Language Clustering

The next step was the use of matrix $\mathbf{M}$ for clustering languages. The Affinity Propagation algorithm [9] was used for clustering, which performs clustering without an a priori specification of the number of clusters; this allows for a natural reflection of language groups without artificially setting the number of clusters. As a result of the algorithm's operation, 6 clusters were formed, which are collected in Table 2. The remaining languages, due to their unique morphology, were treated as isolated languages.

The Affinity Propagation algorithm correctly clustered the languages $\mathcal{L} \in \mathcal{C}$, faithfully reflecting existing language groups. The group {Bosnian, Croatian, Serbian, Slovenian} reflects the South Slavic language group, while the group {Czech, Slovak} corresponds to the West Slavic language group [11]. It is worth noting that the Polish language, despite its genealogical affiliation with Czech and Slovak, does not belong to this group. This results from the unique orthography of the Polish language. Sounds that are written with diacritics in Czech and Slovak (e.g., š, č) are represented by digraphs in Polish (e.g., sz, cz) [12]. The group {Estonian, Finnish} corresponds to the Balto-Finnic group [13], while the group {Indonesian, Malay} is a reflection of the Malay group [14]. The group {Danish, Dutch, German, Norwegian, Swedish} corresponds

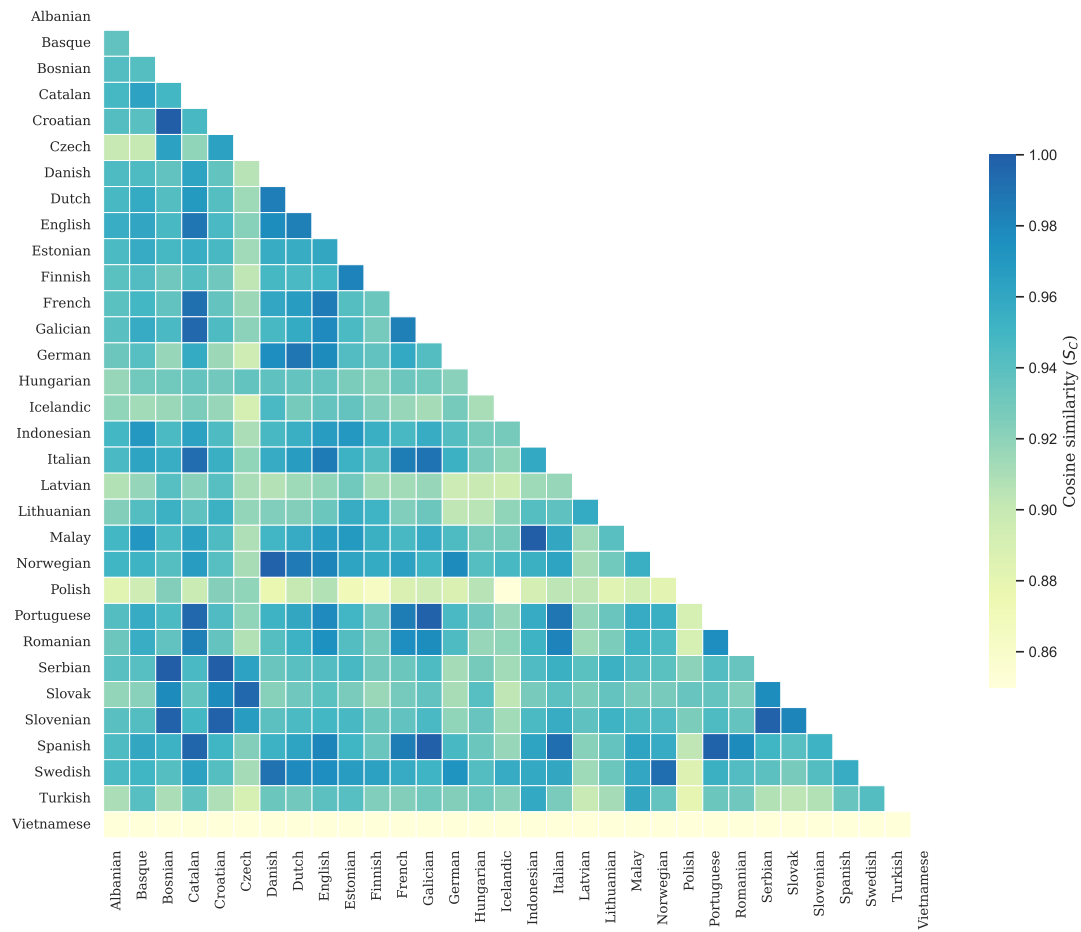


Figure 1: Language correlation map as a graphical representation of matrix M

to both the Germanic and Scandinavian groups, which is a consequence of the high morphological similarity between these groups, caused by years of trade between populations, which prompted a natural transformation of languages towards greater unification [15].

The final group is {Catalan, English, French, Galician, Italian, Portuguese, Romanian, Spanish}, which corresponds to the group of Romance languages. Classifying the English language among Romance languages might appear to be an error; however, despite a different genealogy, the influence of Romance languages on English has significantly altered its morphology, effectively making it structurally similar to other Romance languages [16]. Given a corpus of 15,000 most frequent words, over 50% of the English vocabulary is of Latin and French origin, which significantly outweighs words of Germanic origin (approximately 26%) [17]. This example demonstrates the high effectiveness of the algorithm, which, despite the different origins of languages, is able to indicate their mutual influence on one another.

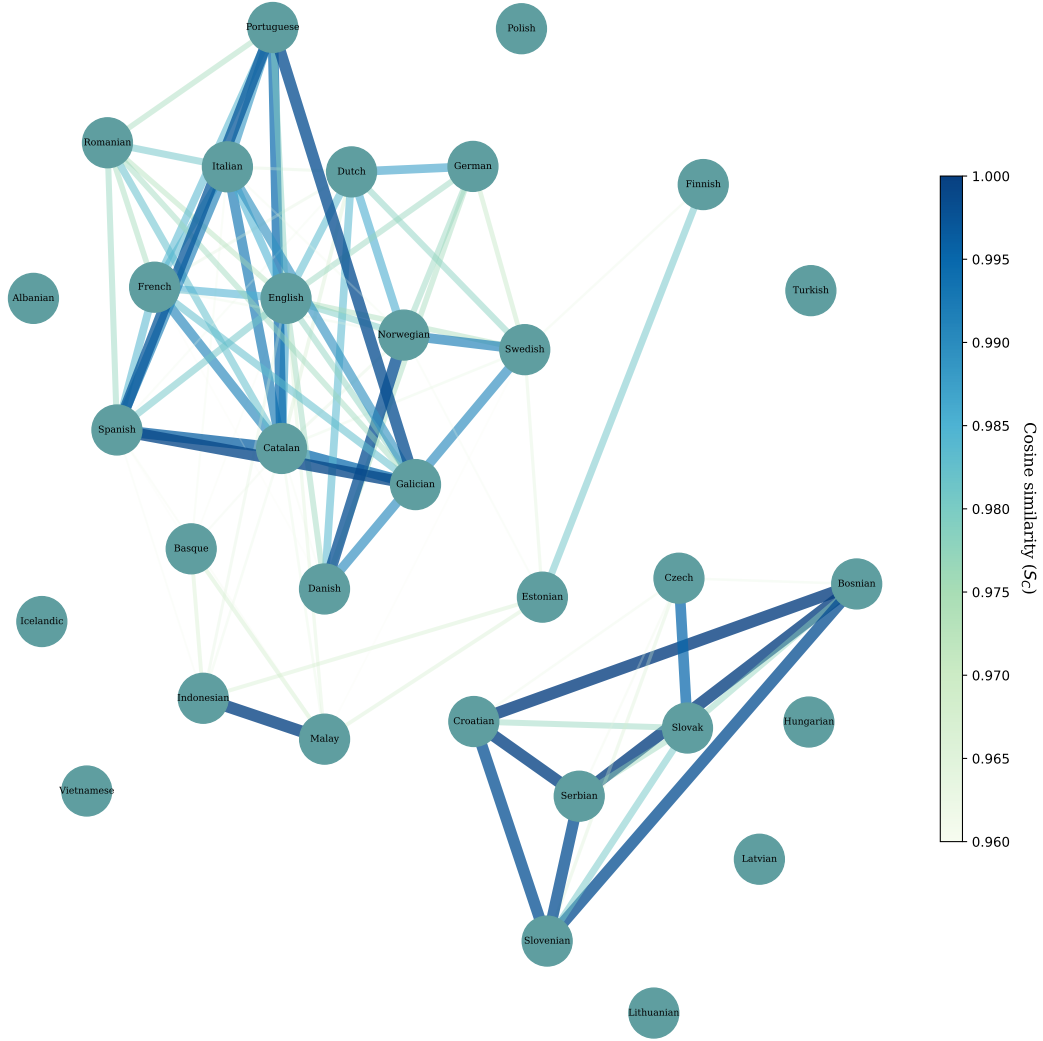


Figure 2: Graph generated based on matrix $\mathbf{M}$ with a threshold of $S_C > 0.96$

3.4. Quantitative Morphological Analysis

The final step of the research was the extraction of quantitative morphological data from each of the six language groups. The morphological characterization was determined according to formulas (5–8) and Algorithm 1. The complete characteristics for each of the six groups are collected in Table 3.

The analysis of the obtained data confirms significant differences between the clusters. Groups exhibiting the highest 2-gram entropy H_2 , such as the Czech cluster ($H_2 = 8.52$ bit) or the Danish cluster ($H_2 = 7.95$ bit), are characterized by high word structure complexity and a

Table 2

Clusters formed as a result of the Affinity Propagation algorithm.

Group $\mathcal{G}$ exemplar	Group members
Bosnian	Bosnian, Croatian, Serbian, Slovenian
Catalan	Catalan, English, French, Galician, Italian, Portuguese, Romanian, Spanish
Czech	Czech, Slovak
Estonian	Estonian, Finnish
Indonesian	Indonesian, Malay
Danish	Danish, Dutch, German, Norwegian, Swedish

Table 3

Morphological characteristics of the identified clusters.

Group $\mathcal{G}$ exemplar	$ \mathcal{G} $	$ \overline{w} $	σ	P_V [%]	$[\overline{w}]_C$	H_2 [bit]
Bosnian	4	6.43	1.99	42.30	1.78	7.82
Catalan	8	6.88	2.26	43.30	1.79	7.86
Czech	2	6.38	1.96	41.50	1.79	8.52
Estonian	2	7.05	2.21	46.20	1.67	7.70
Indonesian	2	6.92	2.56	42.30	1.69	7.34
Danish	5	6.85	2.50	37.40	2.02	7.95

high frequency of rare character combinations. This is directly reflected in the values of the average maximum consonant cluster length $[\overline{w}]_C$, which reaches its highest value ($[\overline{w}]_C = 2.02$) precisely for the Danish cluster. Conversely, clusters with lower 2-gram entropy H_2 , such as the Estonian or Indonesian clusters, are characterized by a higher percentage of vowels P_V , with the Estonian cluster reaching the highest value ($P_V = 46.20\%$).

The final element of the quantitative analysis of the formed groups was the extraction of the most frequent affixes in each group. For this purpose, an analysis of the most frequent boundary 3-grams (containing word boundary markers) was conducted. The results of this analysis for the individual groups are presented in Table 4.

Table 4

Most frequent affixes for the identified language clusters.

Group $\mathcal{G}$ exemplar	Top 5 initial 2-grams	Top 5 terminal 2-grams
Bosnian	<i>po-, pr-, na-, za-, do-</i>	<i>-ti, -la, -te, -je, -na</i>
Catalan	<i>co-, re-, pr-, de-, ca-</i>	<i>-es, -os, -te, -er, -as</i>
Czech	<i>ne-, po-, pr-, za-, vy-</i>	<i>-te, -la, -li, -ní, -ie</i>
Estonian	<i>va-, ka-, ta-, ko-, pa-</i>	<i>-in, -ta, -an, -en, -si</i>
Indonesian	<i>me-, pe-, be-, di-, ke-</i>	<i>-an, -ya, -ng, -ah, -mu</i>
Danish	<i>be-, st-, ve-, ge-, ma-</i>	<i>-en, -er, -et, -de, -te</i>

As the table above shows, the algorithm correctly identified the affixes for the designated language groups. The Indonesian cluster reflects the affixes of Austronesian languages [18],

while suffixes such as *-es*, *-os*, and *-as* in the Catalan cluster directly correspond to the plural endings in Romance languages like Spanish [19]. The prefixes of the Bosnian and Czech clusters reflect the most common Slavic language prefixes [20], whereas the affixes of the Danish cluster derive directly from verb conjugation [21]. Estonian languages do not possess classical prefixes related to word formation [22]; for this reason, the algorithm captured the most frequent stem roots, such as *ka-* and *ta-*, derived from words like *kas* and *tahan*.

4. Conclusion

The study aimed to investigate whether utilizing a simple algorithm based on 3-gram vectorization, employing Word2Vec and Affinity Propagation, would be able to correctly reflect language families.

As the data analysis demonstrated, the algorithm correctly reconstructed the Romance, Germanic, Slavic, and Austronesian families, operating solely on raw text without access to external dictionaries. The model accurately captured orthographic, lexical, and morphological differences; among other things, it extracted the most frequent affixes for the formed language groups, which correspond to the actual affixes of the selected families. A further outcome of the study was the determination of the characteristics collected in Table 3, which provide a summary of the key features of each language group.

One of the primary limitations of the model is that the use of raw text instead of phonetic transcription leads to a high sensitivity to local orthographic features (e.g., the failure to recognize Polish as belonging to the same group as Czech and Slovak). Furthermore, the significant mutual influence between certain languages (as exemplified by English) can cause the lexicon to change so substantially that the algorithm is unable to separate languages of different genealogical origins despite their distinct roots.

The collected characteristics may serve as a foundation for further research. Given the effectiveness of the proposed algorithm, a promising avenue for further development would be to extend the analysis to include languages utilizing different writing systems (e.g., Cyrillic, Greek, or Arabic scripts) and to investigate how corpus size influences the final outcomes.

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